

LEARNING OBJECTIVES.

- *Evaluate Bayesian regression models.*

How fair is the model?

How wrong is the model?

Recall the assumptions of our Normal linear regression model:

Posterior Predictive Check

Consider a regression model with response variable Y , predictor X , and a set of regression parameters θ . Further, suppose that based on a data sample of size n , we simulate an N -length Markov chain $\{\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}\}$ for the posterior model of θ . Then a “good” Bayesian model produces *posterior predicted sets* of n Y values with features similar to the original Y data. To evaluate whether your model satisfies this goal, do the following:

Example 1. Perform a posterior predictive check of the model of Centre GPA based on absences from Handout 11. Summarize your findings.

Dealing With Wrong Models

How accurate are the posterior predictive models?

Predicting Original Data Points

How good of a job does our model do at predicting the values of a data point used to build the model? We can answer this question in several ways:

We can then evaluate the overall model fit using the following:

Example 2. Examine the posterior predictive summaries and a plot of the posterior predictive intervals for our model. Summarize the results.

Cross-Validation

The above results can be misleading, because we are using the same points to build and evaluate the model. We can instead use cross-validation, in which we train and test the model on different subsets of our data. A common way to do this is with **k-fold cross-validation**:

Example 3. Report posterior predictive summaries for our GPA model using 10-fold cross-validation. Compare the results to those produced earlier.

Example 4. Consider the model from Example 7 of Handout 11. Perform a posterior predictive check to determine if the assumptions of our model are reasonable. Then report posterior predictive summaries from both the full data set and 10-fold cross-validation.